ORG
MULTI AGENTIC SIMULATION GAME

225522M

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CM3630 - MULTI AGENT SYSTEMS

BSC HONS (AI)

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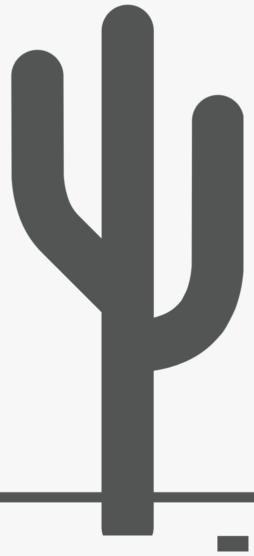


ISSUES

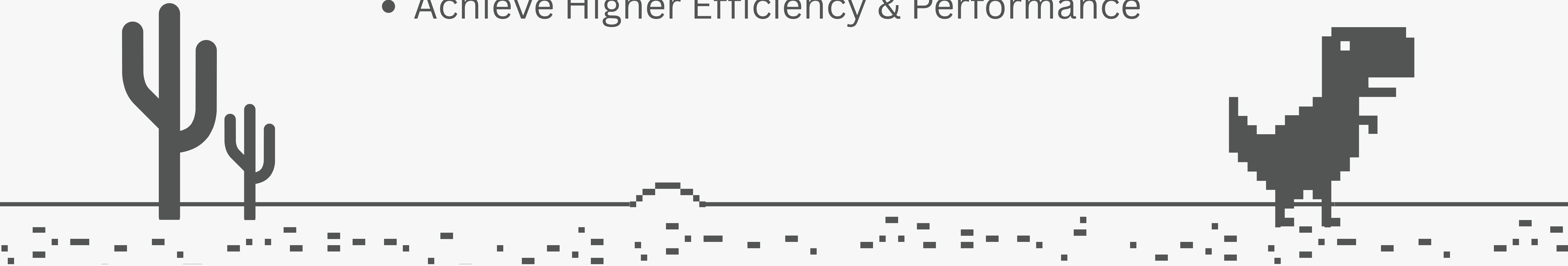
The NPC opponent team does not operate as a Multi-Agent System. No communication between team members

Leads to inefficiency and slow decision-making.

SOLUTION

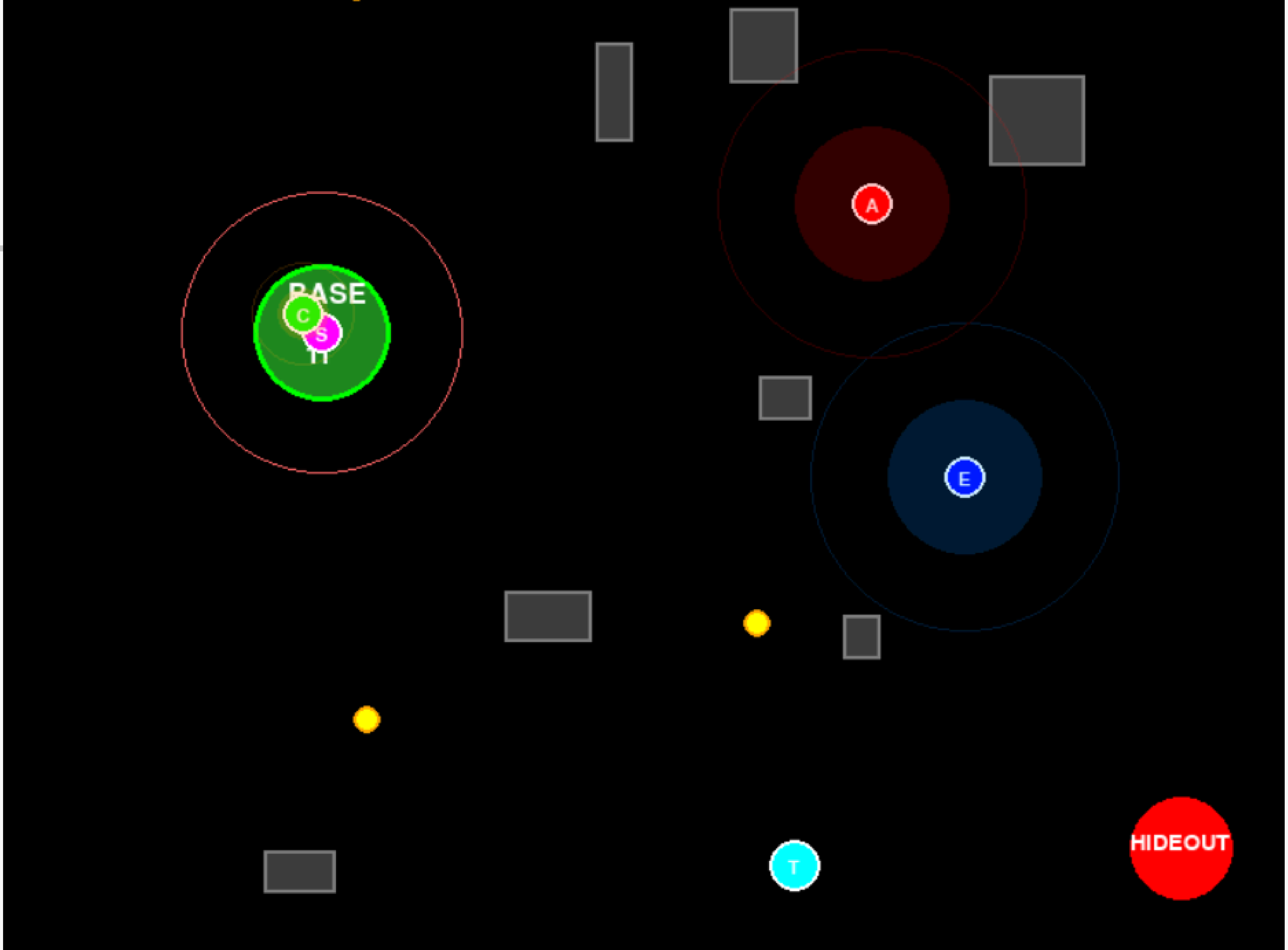
Implementing a Multi-Agent System for opponent team.

- Enable Communication Between Agents
- Coordinate Roles & Responsibilities
- Use Distributed Decision-Making
- Achieve Higher Efficiency & Performance



AGENTS

- Attacker - Pursues and attempts to capture the thief.
- Collector - Gathers resources and delivers them to base camp.
- Explorer - Scouts the map to discover resources and detect the thief.
- Strategist - Coordinates all agents and makes strategic decisions.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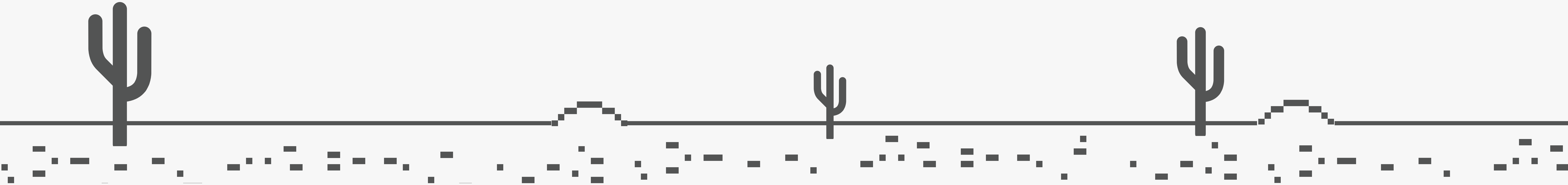
MAS FEATURES

Communication - peer-to-peer, broadcast

Coordination - Hierarchical

AGENT FEATURES

- Adaptivity
- Situatedness
- Sociability



TECHNOLOGY OVERVIEW

Programming Framework:

- **MESA for agent-based modeling and simulation**

Visualization Framework:

- **Pygame**

DEMOSTRATION


GAME OVER